

Experiment No: 5

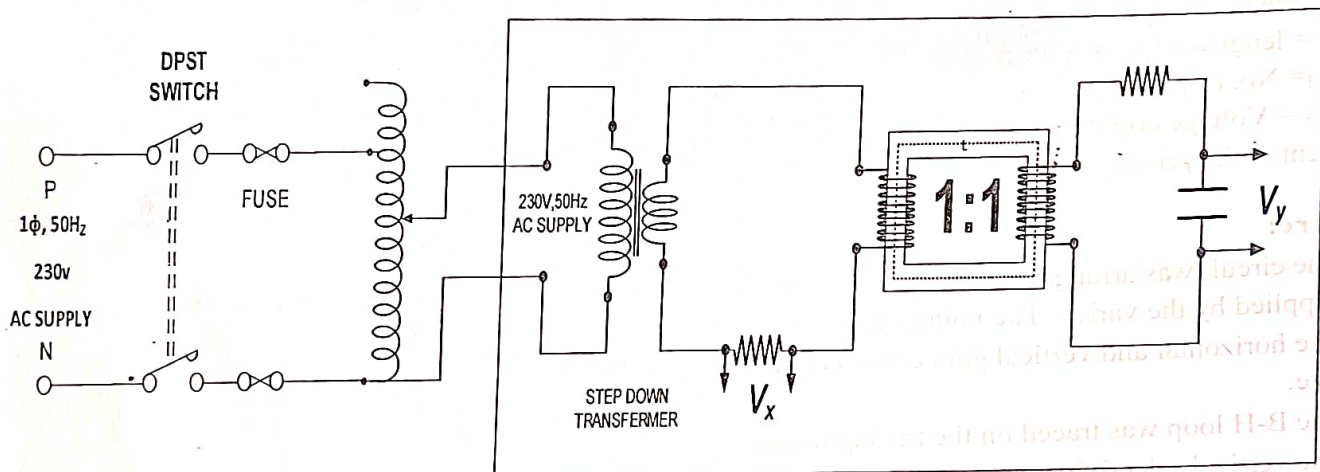
Measurement of Hysteresis Loss

Aim of the experiment: To trace Hysteresis loop of a specimen and find the Hysteresis loss.

Apparatus Required:

Sl.no.	Name of Item	Rating	Type	Quantity
1.	B-H Curve trainer			1 No.
2.	variac			1 No.
3.	CRO	30 MHz	Dual trace	1 No.
4.	CRO Probes	1:10, 1:1		one each
5.	Connecting Wires	1.5 mm ²		As reqd.

Circuit Diagram:



Theory:

Magnetic Induction B is measured in terms of voltage " V_y " across the integrating capacitor

$$\begin{aligned} V_y &= \frac{1}{C} \int i_2 dt \\ &= \frac{1}{C} \int \frac{\varepsilon_2}{Z_2} dt \\ &= \frac{1}{CR_2} \int \varepsilon_2 dt \\ &= \frac{N_2}{CR_2} \int \left| -\frac{d\phi}{dt} \right| dt \end{aligned}$$

(Since, ε_2 : induced emf in the secondary, $Z_2 =$

$$\sqrt{R_2^2 + (X_L - X_C)^2} \text{ and } X_L \approx X_C$$

$$= \frac{N_2 \phi}{CR_2}$$

$$V_y = \frac{N_2 BA}{CR_2}$$

$$B \propto V_y$$

$$\text{Magnetic field } H = \frac{N_1}{L} i_1 = \left(\frac{N_1}{L R_1} \right) V_x$$

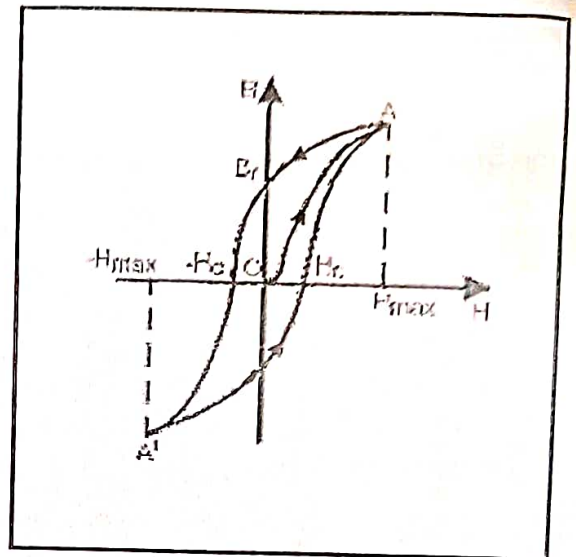
$$H \propto V_x \text{ where,}$$

L = length of the specimen in meter

N_1 = No. of primary turns.

V_x = Voltage applied to horizontal plane.

i_1 = Current in the primary coil



Procedure:

- The circuit was arranged as shown in the circuit diagram the input to primary of the core was supplied by the variac. The voltage across C_2 feeds the vertical input of the CRO.
- The horizontal and vertical gain controls of the CRO were adjusted to get a loop of the convenient size.
- The B-H loop was traced on the tracing paper.
- The vertical selectivity (S_1) and the horizontal selectivity (S_2) were determined. The sensitivity was expressed in volts per centimeter.
- Flux density:- $B = (R_2 C / N_2 A) V_y$ Wb/m²
- Hysteresis loss:- $\vec{B} \cdot \vec{H} = (R_2 C / N_2 A) (N_1 / R_1 L) V_x \cdot V_y$ Joule/m³/cycle

Calculation:

The physical quantities: $B_{\max}$, Retentivity (B_r), Coercivity (H_c) and Hysteresis loss ($\vec{B} \cdot \vec{H}$) per volume per cycle are calculated from the data corresponding to different positions on the hysteresis loop observed in the oscilloscope screen and traced in the graph sheet.

The hysteresis loss is calculated as follows:

The horizontal scale of the oscilloscope is adjusted to read one volt per division which is equivalent to 1 cm in the graph sheet. One small division in the graph sheet therefore is equivalent to 0.1 V

The vertical scale oscilloscope is adjusted to read 50 mV per division which equivalent to 1 cm in the graph sheet. One small division in the graph sheet, therefore is equivalent to 5.0×10^{-3} V.

The area of each small division in the graph sheet $v_x v_y = 5.0 \times 10^{-4} \text{ V}^2$

If number of small divisions enclosed in the B-H loop on the graph sheet is n , the equivalent quantity:

$$V_x V_y = \sum v_x v_y = n \times 5.0 \times 10^{-4} \text{ V}^2 \text{ from which the hysteresis loss is calculated as } \vec{B} \cdot \vec{H} =$$

$$\frac{R_2 C}{N_2 A} \times \frac{N_1}{R_1 L} \times n \times 5.0 \times 10^{-4} \frac{\text{J}}{\text{m}^3} / \text{cycle}$$

Conclusion:

The physical quantities $B_{\max}$, Retentivity, Coercivity and hysteresis loss are found to be -----, -----, ----- and -----, respectively